CANONICAL DIVIDED POWER STRUCTURE ON TOR OVER A COMMUTATIVE RING

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Editorial note. Declared dependency: lemma-dpdga-good is not carded (its span omits the verification that the descended maps form a divided power structure); recorded via external_references, not re-proved.

Allowed standard prerequisites (editorial).

- Tate resolution. lemma-tate-resolution (same chapter, named). Supplies the factorization R -> A -> S computing Tor.
- Base change of DP structures. lemma-base-change-div (same chapter, named). Endows B = A (x)_R T and A[y_j] with compatible DP structures.
- Descent to homology. lemma-dpdga-good (same chapter, named; NOT carded – build-gate exclusion). Descends the DP structure from the DGA to its homology; declared as an external reference, not re-proved (its own conclusion-verification is omitted in the source).
- Independence of the resolution. lemma-uniqueness-tate-resolution (same chapter, named). Compares two choices of A by a structure-preserving quasi-isomorphism.

Source reference. Original lemma 09PL: sources/stacks-dpa/pinned-original-source.tex, source lines 1365–1365. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 09PK: sources/stacks-dpa/pinned-original-source.tex, source lines 1320–1320. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 09PP: sources/stacks-dpa/pinned-original-source.tex, source lines 1546–1546. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let R be a commutative ring. Let S and T be commutative R-algebras. Then there is a canonical structure of a strictly graded commutative R-algebra with divided powers on*

$$\mathrm{Tor}_*^R(S, T).$$

Date: stacks-project:dpa.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Choose a factorization $R \to A \to S$ as above. Since $A \to S$ is a quasi-isomorphism and since A_d is a free R-module, we see that the differential graded algebra $B = A \otimes_R T$ computes the Tor groups displayed in the lemma. Choose a surjection $R[y_j : j \in J] \to T$. Then we see that B is a quotient of the differential graded algebra $A[y_j : j \in J]$ whose homology sits in degree 0 (it is equal to $S[y_j : j \in J]$). By Lemma 09PL the differential graded algebras B and $A[y_j : j \in J]$ have divided power structures compatible with the differentials. Hence we obtain our divided power structure on $H(B)$ by Lemma 09PK.

The divided power algebra structure constructed in this way is independent of the choice of A. Namely, if A' is a second choice, then Lemma 09PP implies there is a map $A \to A'$ preserving all structure and the augmentations towards S. Then the induced map $B = A \otimes_R T \to A' \otimes_R T' = B'$ also preserves all structure and is a quasi-isomorphism. The induced isomorphism of Tor algebras is therefore compatible with products and divided powers. □